ModelIt!

Teacher's Guide

Can We Edit the Human Genome?

Modeling CRISPR Gene Editing, Inheritance, and the Ethics of Genetic Engineering

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-LS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS3-1

Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring.

Three-Dimensional Learning

Science Practice	**Asking Questions and Defining Problems** — Students ask clarifying questions about the relationships between DNA structure, gene function, and how CRISPR-mediated changes to DNA sequences affect protein production and trait expression.
Disciplinary Core Idea	**LS3.A Inheritance of Traits** — DNA sequences code for proteins that determine traits; CRISPR allows modification of these sequences, and whether changes are in somatic or germline cells determines whether modifications are inherited by offspring.
Crosscutting Concept	**Structure and Function** — Students connect the molecular structure of DNA, guide RNA, and Cas9 protein to the function of precise gene editing, understanding how structural specificity determines functional accuracy.

CAST Testing Alignment

MULTIPLE CHOICE:

A CRISPR guide RNA with 99.8% specificity is used to edit embryonic cells. Given that the human genome contains 3.2 billion base pairs, approximately how many potential off-target sites could be affected, and why does this matter for germline editing?

CONSTRUCTED RESPONSE:

Using your model, compare the predicted Therapeutic Outcome for editing a single gene causing cystic fibrosis versus editing the 20+ genes associated with height. Explain why the model predicts dramatically different outcomes and what this reveals about the limits of gene editing for complex traits.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

CRISPR-Cas9 is arguably the most revolutionary biological technology since the discovery of DNA's structure. Adapted from a natural immune system that bacteria use to defend against viruses, CRISPR allows scientists to edit any organism's DNA with a precision that was unimaginable a decade ago. The technology is transforming medicine, agriculture, and our fundamental understanding of genetics — while raising ethical questions that humanity has never faced before.

How CRISPR Works

The CRISPR-Cas9 system has two key components: a guide RNA (gRNA) that is designed to match a specific 20-nucleotide DNA target sequence, and the Cas9 protein that acts as molecular scissors. The gRNA leads Cas9 to the correct location in the genome, where Cas9 cuts both strands of the DNA double helix. The cell's natural repair mechanisms then fix the break — either by joining the cut ends (often disabling the gene) or by using a provided DNA template to insert a new sequence. This allows scientists to delete harmful mutations, correct disease-causing variants, or insert entirely new genetic sequences.

Current Medical Applications

CRISPR has already produced FDA-approved therapies. Casgevy (2023) treats sickle cell disease and transfusion-dependent beta-thalassemia by editing patients' own blood stem cells to reactivate fetal hemoglobin production. Clinical trials are underway for CRISPR treatments of cancer (editing immune cells to better attack tumors), hereditary blindness (Leber congenital amaurosis), HIV (disabling the CCR5 receptor), and dozens of other genetic conditions. Most current applications use somatic cell editing — modifying the patient's cells without affecting their children.

The Germline Editing Controversy

In 2018, Chinese scientist He Jiankui shocked the world by announcing he had used CRISPR to edit human embryos — twin girls born with modified CCR5 genes intended to confer HIV resistance. The scientific community condemned the experiment as premature, poorly designed, and ethically unjustifiable. The girls may face unknown health consequences from off-target effects. The incident crystallized the debate: germline edits are permanent, heritable, and affect individuals who cannot consent. Most countries have banned germline editing, but enforcement is inconsistent.

The Enhancement vs. Therapy Debate

The most contentious ethical frontier is the line between therapy (fixing a disease-causing gene) and enhancement (improving a normal trait). Curing Huntington's disease seems clearly therapeutic. But what about editing genes associated with higher intelligence, greater muscle mass, or longer lifespan? Most traits are polygenic — influenced by dozens or hundreds of genes interacting with environment — making enhancement both technically difficult and ethically fraught. Critics warn of a new eugenics where wealthy families purchase genetic advantages, creating biological inequality between economic classes.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify guide RNA specificity, target gene location, editing efficiency, off-target mutation rate, and therapeutic outcome as the key components of the CRISPR gene-editing system.

E **Establish Relationships**

Students determine that guide RNA specificity is the primary determinant of editing accuracy, that off-target effects increase with lower specificity and multiple targets, and that therapeutic outcome depends on the balance of on-target success and off-target harm.

V **Visualize & Model**

Students build a computational model showing how guide RNA design, target complexity, and cellular repair mechanisms interact to determine the safety and effectiveness of gene edits.

E **Evaluate Outcomes**

Students run single-gene, multi-gene, and germline editing scenarios to test predictions about when CRISPR is safe and effective versus when risks outweigh benefits.

R **Revise & Extend**

Students add delivery method efficiency, mosaicism risk, or epigenetic disruption to explore how real-world biological complexity affects gene-editing outcomes beyond the simplified model.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"What if you could rewrite the code of life itself? Today we investigate the most powerful biological technology ever created."*

Watch For

Show cover image. Ask: What have you heard about CRISPR or gene editing? Quick share-out.

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll be able to do by the end of today — model how CRISPR edits DNA and evaluate when it should be used."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with molecular animation or diagram of CRISPR mechanism.

Slide 3: Big Question

3 min

Say: *"CRISPR can cure sickle cell disease AND theoretically enhance intelligence. Should we use it for both? Neither? Where's the line?"*

Watch For

Quick-write: Students write their initial position on where the line should be drawn. Save to revisit after modeling.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model the actual molecular mechanism of CRISPR to understand both its power and its limitations."*

Watch For

Preview each LEVER step. Emphasize that understanding the biology is essential for informed ethical reasoning.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this gene-editing system? What inputs do scientists control, and what emerges from the biology?"*

Watch For

Guide sorting of external vs. internal components. Discuss: What can scientists design, and what does biology determine?

Slide 6: Activity 2: Connections

8 min

Say: *"If Guide RNA Specificity decreases even slightly, what happens to Off-Target Mutation Rate across 3.2 billion base pairs?"*

Watch For

Students draw arrows showing +/- relationships. Discuss why the genome's massive size amplifies even tiny error rates.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run each scenario. Compare what happens when you edit one gene precisely vs. multiple genes simultaneously."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss why complex traits are so much harder to edit safely.

Slide 8: Discoveries

5 min

Say: *"Our model reveals both the extraordinary promise and the real limitations of gene editing technology."*

Watch For

Class discussion of discoveries. Revisit initial positions — has the model evidence changed anyone's thinking about where the line should be?

Slide 9: STEM Challenge

12 min

Say: *"Now apply both the science AND the ethics. Your hospital needs a framework for deciding which gene edits to*

approve."

Watch For

Groups design ethics frameworks using model data to set scientific criteria. Present and debate frameworks.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Guide RNA Specificity, Target Gene Location
- **INTERNAL:** Editing Efficiency, Off-Target Mutation Rate, Therapeutic Outcome

Relationships

- **Guide RNA Specificity to Editing Efficiency:** POSITIVE (+) — Higher guide RNA specificity means the CRISPR system binds more reliably to the intended target, increasing the percentage of cells where the correct edit is made.
- **Guide RNA Specificity to Off-Target Mutation Rate:** NEGATIVE (-) — Higher specificity reduces the chance of the guide RNA binding to similar but incorrect DNA sequences elsewhere in the genome, decreasing off-target mutations.
- **Off-Target Mutation Rate to Therapeutic Outcome:** NEGATIVE (-) — More off-target mutations increase the risk of harmful unintended effects — including potential activation of oncogenes — which decreases overall therapeutic benefit.
- **Editing Efficiency to Therapeutic Outcome:** POSITIVE (+) — Higher editing efficiency means more cells carry the corrected gene, producing more functional protein and greater therapeutic benefit for the patient.

Key Discoveries

- CRISPR is highly effective for single-gene disorders like sickle cell disease where one specific mutation can be precisely targeted with well-designed guide RNA
- Off-target effects are the critical safety limitation — even with 99.9% specificity, the sheer size of the genome means unintended cuts are statistically inevitable across millions of cells
- Editing multiple genes simultaneously for complex traits multiplies both the difficulty and the risk — each additional target increases the probability of harmful off-target effects
- Germline editing raises unique concerns because edits are permanent and heritable — mistakes cannot be undone and will propagate through all future generations of that lineage

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"CRISPR can edit any trait we want with perfect precision"	While CRISPR is remarkably precise compared to previous gene-editing tools, it is not perfect. Off-target effects occur because the guide RNA can bind to similar DNA sequences elsewhere in the genome. More importantly, most human traits (intelligence, height, personality) are influenced by hundreds of genes interacting with environment, making them far beyond current editing capabilities. CRISPR works best for single-gene disorders with well-characterized mutations.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Gene editing is the same as GMOs in food"	While both involve modifying DNA, the techniques, organisms, and ethical stakes are fundamentally different. Agricultural GMOs typically insert genes from other species into plants for pest resistance or nutrition (like Bt corn or Golden Rice). Human gene editing with CRISPR corrects or modifies existing human genes. The ethical considerations — informed consent, heritability, equity, human dignity — are unique to human germline editing and don't apply to crop modification.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If a gene edit cures a disease, there are no downsides"	The model clearly shows that every gene edit carries some off-target risk. Additionally, some disease-causing genes confer benefits in other contexts — the sickle cell gene provides malaria resistance, and some cancer-risk genes enhance immune function. Editing these genes may cure one condition while creating vulnerability to another. The concept of pleiotropy (one gene affecting multiple traits) means the full consequences of any edit may not be apparent for years or generations.